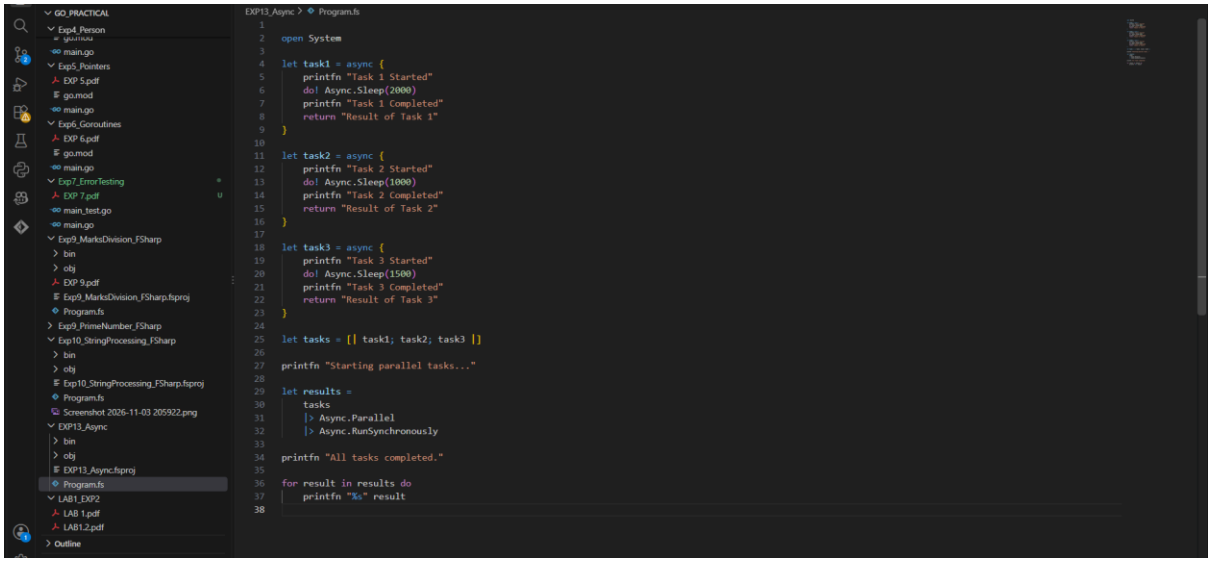
LAB 7

1. Asynchronous and Parallel Workflows in F# (Exp 13)

Write an F# program that

1. uses asynchronous workflows (async { } blocks) to run at least three tasks concurrently using Async.Parallel or Async.StartChild.
2. Tasks should simulate I/O-bound or compute-intensive work (e.g., use Async.Sleep to simulate delay).
3. Collect and print all results after parallel execution.
4. Add output messages demonstrating non-sequential execution.



```
open System

let task1 = async {
    printfn "Task 1 Started"
    do! Async.Sleep(2000)
    printfn "Task 1 Completed"
    return "Result of Task 1"
}

let task2 = async {
    printfn "Task 2 Started"
    do! Async.Sleep(1000)
    printfn "Task 2 Completed"
    return "Result of Task 2"
}

let task3 = async {
    printfn "Task 3 Started"
    do! Async.Sleep(1500)
    printfn "Task 3 Completed"
    return "Result of Task 3"
}

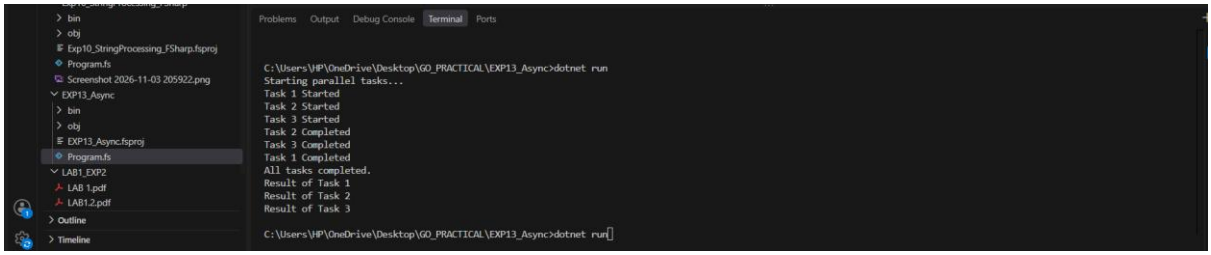
let tasks = [| task1; task2; task3 |]

printfn "Starting parallel tasks..."

let results =
    tasks
    |> Async.Parallel
    |> Async.RunSynchronously

printfn "All tasks completed."

for result in results do
    printfn "%s" result
```



```
C:\Users\HP\OneDrive\Desktop\GO_PRACTICAL\EXP13_Async>dotnet run
Starting parallel tasks...
Task 1 Started
Task 2 Started
Task 3 Started
Task 2 Completed
Task 3 Completed
Task 1 Completed
All tasks completed.
Result of Task 1
Result of Task 2
Result of Task 3

C:\Users\HP\OneDrive\Desktop\GO_PRACTICAL\EXP13_Async>dotnet run
```